11.1.

Period : $f(x) = f(x+p)$

1. $\cos x = \cos(x+2\pi) \qquad p = 2\pi$

$\sin x = \sin(x+2\pi) \qquad p = 2\pi$

$\cos 2x = \cos(2x+2\pi) = \cos(2(x+\pi)) \qquad p = \pi$

$\sin 2x = \sin(2x+2\pi) = \sin(2(x+\pi)) \qquad p = \pi$

$\cos \pi x = \cos(\pi x + 2\pi) = \cos(\pi(x+2)) \qquad p = 2$

$\sin \pi x = \sin(\pi x + 2\pi) = \sin(\pi(x+2)) \qquad p = 2$

$\cos 2\pi x = \cos(2\pi x + 2\pi) = \cos(2\pi(x+1)) \qquad p = 1$

$\sin 2\pi x = \qquad = \sin(2\pi(x+1)) \qquad p = 1$

2. $\cos nx = \cos(nx+2\pi) = \cos\left(n\left(x+\frac{2\pi}{n}\right)\right) \qquad p = \frac{2\pi}{n}$

$\sin nx$ —"— $\qquad$ —"—

$\cos \frac{2\pi x}{k} = \cos\left(\frac{2\pi x}{k}+2\pi\right) = \cos\left(\frac{2\pi}{k}(x+k)\right) \qquad p = k$

$\sin \frac{2\pi x}{k} \qquad$ —"—

$\cos \frac{2\pi n x}{k} = \cos\left(\frac{2\pi n x}{k}+2\pi\right) = \cos\left(\frac{2\pi n}{k}\left(x+\frac{k}{n}\right)\right)$

$p = \frac{k}{n}$

$\sin \frac{2\pi n x}{k} \qquad$ —"—

5 min

3.) $f(x) = f(x+p)$

$g(x) = g(x+p)$

$h(x) = a\,f(x) + b\,g(x) = a\,f(x+p) + b\,g(x+p) =$

$= h(x+p)$ ✓

vector space:

- constant multiple
- addition

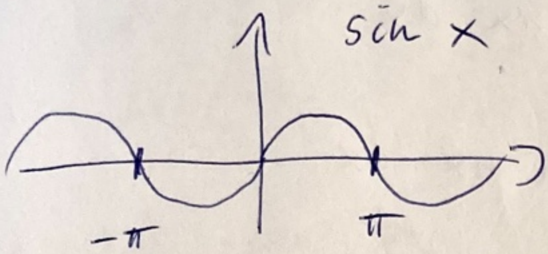
4.) $f(x) = f(x+p)$

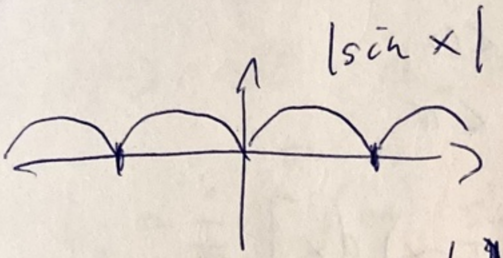
a) $f(ax) = f(ax+p) = f\left(a\left(x+\frac{p}{a}\right)\right) \qquad p' = \frac{p}{a}$

b) $f\left(\frac{x}{b}\right) = f\left(\frac{x}{b}+p\right) = f\left(\frac{x+bp}{b}\right) \qquad p' = bp$

3 min

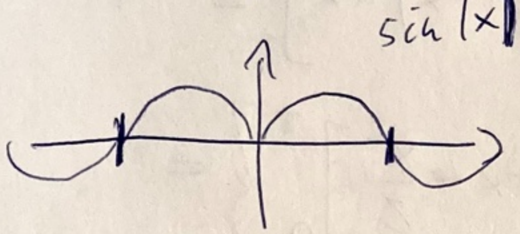
7.) $f(x) = |\sin x|$



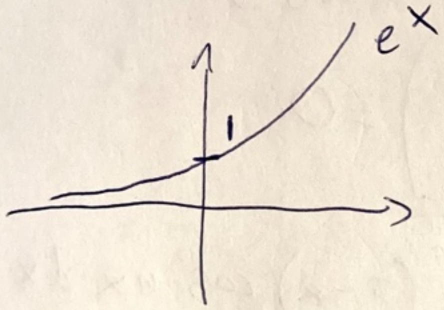



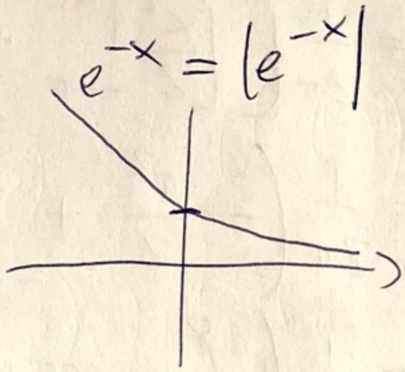

$|\sin x| = \sin|x|$

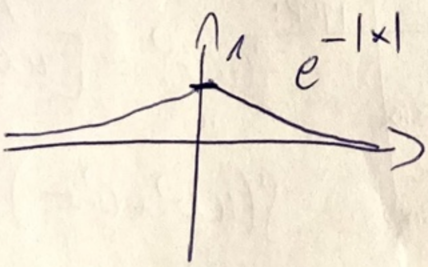
for $-\pi \leq x \leq \pi$



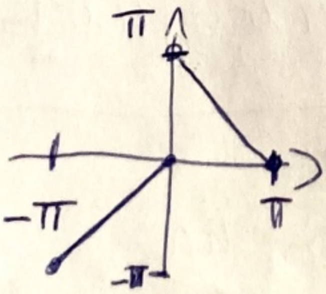

8.)



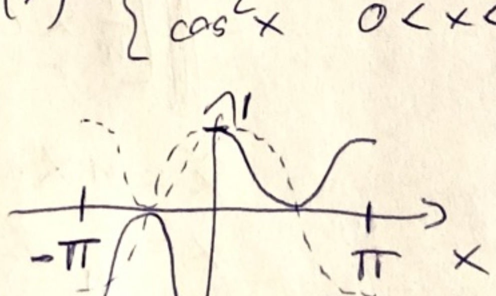




9.)




$$f(x) = \begin{cases} x & -\pi < x < 0 \\ \pi - x & 0 < x < \pi \end{cases}$$

10.) $$f(x) = \begin{cases} -\cos^2 x & -\pi < x < 0 \\ \cos^2 x & 0 < x < \pi \end{cases}$$

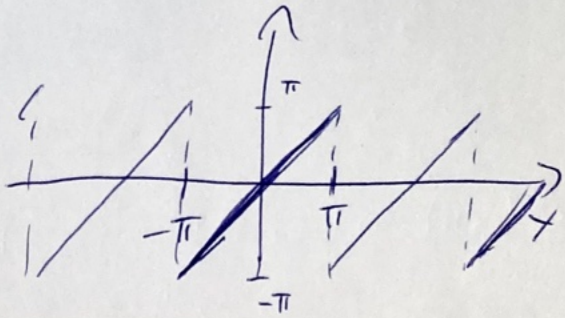



5 min

# Sawtooth wave (11.2 Example 5)

$f(x) = x \qquad -\pi < x < \pi$

$f(x+2\pi) = f(x)$




- odd function

$\Rightarrow$ all $a_n = 0 \qquad n = 0, 1, \ldots$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

- $\int x \sin nx\,dx =$

$= \int \frac{y}{n}\sin y\,\frac{dy}{n} =$

$= \frac{1}{n^2}\int y \sin y\,dy =$

substitution:

$y = nx$

$x = \frac{y}{n}$

$dx = \frac{dy}{n}$

integration by parts:

$\int (uv)' = uv$

$\int (u'v + uv') = uv$

$\int uv' = uv - \int u'v$

$u = y \qquad u' = 1$

$v' = \sin y \qquad v = -\cos y$

$= \frac{1}{n^2}\left(-y\cos y + \sin y\right) =$

$= \frac{1}{n^2}\left(-nx\cos nx + \sin nx\right) = -\frac{1}{n}x\cos nx + \frac{1}{n^2}\sin nx$

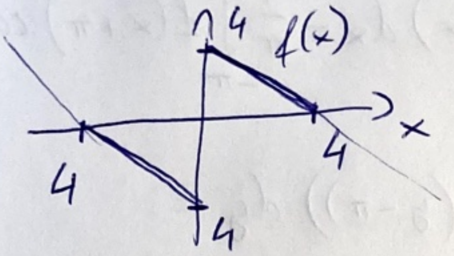
$b_n = \frac{1}{\pi}\left[-\frac{1}{n}x\cos nx + \frac{1}{n^2}\sin nx\right]_{-\pi}^{\pi} =$

$\cos n\pi = (-1)^n$

$\sin n\pi = 0$

$= \frac{1}{\pi}\left(-\frac{1}{n}\pi(-1)^n + 0 - \frac{1}{n}\pi(-1)^n - 0\right) = \underline{\underline{-\frac{2}{n}(-1)^n}}$

11.2.10) Find the Fourier series without calculating integrals, reuse the previous result for sawtooth.




- Change of period

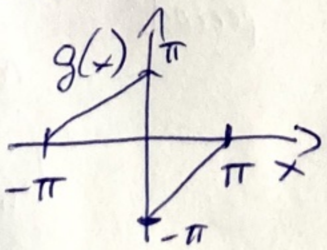
$f(x) = f(x+2L)$ 2L-periodic

let $g(x) = f\left(\frac{L}{\pi}x\right)$ $2\pi$-periodic

Fourier ~~series~~ coefficients of f with 2L period are the same as of g with $2\pi$ period

- Linearity: multiplying f with a constant multiplies the Fourier coefficients. Adding two functions adds the coeffs.

Let's look at




coefficients for g $\rightarrow$ multiply by $-\frac{4}{\pi}$ $\rightarrow$ coeffs for f.

- $\pi$ - shift (or half period shift)

$$g(x) = f(x+\pi)$$

$$a_n' = \frac{1}{\pi}\int_{-\pi}^{\pi} g(x)\cos(nx)\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x+\pi)\cos(nx)\,dx =$$

$y = x+\pi$

$$= \frac{1}{\pi}\int_{0}^{2\pi} f(y)\cos(n(y-\pi))\,dy =$$

periodicity

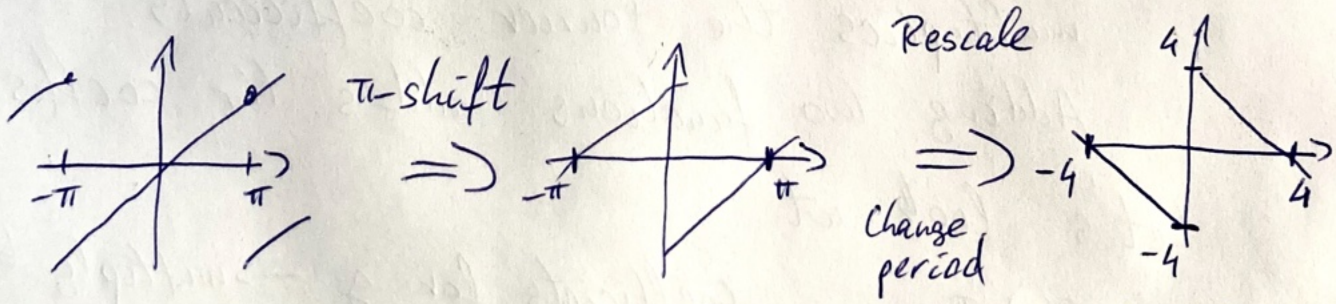
$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(y)\cos(ny - n\pi)\,dy =$$

$$\cos(x+\pi) = -\cos(x)$$
$$\cos(x+2\pi) = \cos(x)$$
$$\Rightarrow \cos(x+n\pi) = \cos(x)\cdot(-1)^n$$

$$= (-1)^n \frac{1}{\pi}\int_{-\pi}^{\pi} f(y)\cos(ny)\,dy = (-1)^n a_n$$

Same for sine terms

$$b_n' = (-1)^n b_n$$




$a_n = 0$ $\quad$ $b_n = -\frac{2}{n}(-1)^n$

$a_n = 0$ $\quad$ $b_n = -\frac{2}{n}$

multiply by -4/π

$a_n = 0$

$$b_n = \frac{4}{\pi}\cdot\frac{2}{n}$$

$$\underline{\underline{b_n = \frac{8}{\pi n}}}$$